

Generative Nonlinear Observer for Indistinguishable Systems (long version)

Johan Peralez, Clara L. Galimberti, Madiha Nadri

Abstract—Nonlinear state estimation under indistinguishability consideration or significant measurement noise remains a challenging problem. While the Kazantzis-Kravaris-Luenberger (KKL) observer offers a robust framework by immersing nonlinear dynamics into a stable linear filter, reconstructing the physical state from the latent space is often ill-posed. In singular regions, the transformation is non-injective, and standard neural regression yields physically inconsistent mean estimates. In noisy scenarios, exact inversion amplifies measurement errors.

In this paper, we propose *KKL-CFM*, a novel architecture coupling linear KKL dynamics with Conditional Flow Matching (CFM). Instead of learning a deterministic inverse, we approximate the conditional distribution of the state. This generative approach allows for (i) capturing complex multimodal distributions when observability is lost, and (ii) empirical resilience to noise in stochastic environments. Furthermore, we introduce a *Clustering-and-Tracking* scheme to track the system’s distinct physical branches over time.

I. INTRODUCTION

State observation for nonlinear systems is a fundamental problem in control theory. Among geometric approaches, the Kazantzis-Kravaris-Luenberger (KKL) observer [1]–[3] is particularly attractive as it allows immersing nonlinear dynamics into a stable linear filter. However, its practical applicability relies on the existence and computability of a left-inverse mapping from the latent space back to the physical state.

Recent approaches [4]–[6] successfully approximate this inversion using deep learning. While effective for fully observable deterministic systems, these regression-based methods face severe limitations when the system is not backward-distinguishable. Indeed, when observability is singular, a case frequently encountered in induction motors [7] and permanent magnet synchronous motors [8], the underlying mapping is non-injective. In such scenarios, minimizing the mean squared error forces the network to converge toward the arithmetic mean of all indistinguishable states, often resulting in physically impossible state reconstructions.

While traditional local approaches (e.g., Extended Kalman Filters) and High-Gain Observers fail to capture multiple valid trajectories or require strict uniform observability, stochastic methods like Particle Filters theoretically maintain multiple hypotheses but often suffer from heavy computational burdens and particle weight degeneracy [9], [10].

A major theoretical breakthrough was recently achieved with the introduction of set-valued KKL immersions [11], which formally establishes that the observer state maps to the entire set of indistinguishable physical trajectories rather than a single point. While this conceptual framework is foundational,

its numerical deployment relies on optimization-based inversion techniques that are computationally prohibitive, highly sensitive to measurement noise, and often collapse to a single local minimizer in practice. To overcome these practical bottlenecks, we introduce KKL-CFM as a scalable computational architecture bridging these theoretical set-valued immersions with deep generative modeling.

Our main contributions are:

- *Generative KKL Inversion*: We replace deterministic regression with continuous normalizing flows to learn the state distribution given the observer’s latent state. This successfully captures all indistinguishable physical states when observability is lost.
- *Trajectory Reconstruction*: We propose a Clustering-and-Tracking scheme to isolate and follow distinct physical branches over time, avoiding unstable jumps or physically meaningless state averaging.

II. PROBLEM STATEMENT

Consider the continuous-time nonlinear system:

$$\dot{x} = f(x), \quad y = h(x) + v, \quad (1)$$

where $x \in \mathbb{R}^{n_x}$ is the state constrained to a compact set $\mathcal{X}$, $y \in \mathbb{R}^{n_y}$ is the measured output, and $v \in \mathbb{R}^{n_v}$ is an unknown measurement noise. The vector field f is locally Lipschitz and h is continuous.

A. Background: The KKL Observer Framework

To estimate the state from the output history, we adopt the KKL observer framework [1], [2]. It associates the disturbance-free system ($v = 0$) with a linear latent dynamical system:

$$\dot{z} = Az + By, \quad (2)$$

where $z \in \mathbb{R}^{n_z}$ with the latent dimension typically chosen as $n_z \geq (2n_x + 1)n_y$ to ensure immersion properties. The matrix A is Hurwitz, and (A, B) is a controllable pair. Under mild assumptions, there exists a mapping $T : \mathcal{X} \rightarrow \mathbb{R}^{n_z}$ satisfying the PDE $\frac{\partial T}{\partial x}(x)f(x) = AT(x) + Bh(x)$. Consequently, the latent state $z(t)$ converges exponentially to $T(x(t))$.

However, the classical KKL framework assumes a deterministic, single-valued inverse mapping. This standard approach becomes fundamentally ill-posed under the system’s indistinguishability, where a single latent history maps to multiple valid physical states.

In our setting, the presence of the measurement noise v in (1) inherently stochasticizes the latent state. In the particular

case where v is modeled as a Gaussian white noise with covariance Q_v , the observer dynamics (2) become an Itô Stochastic Differential Equation, and the probability density $p(z, t)$ diffuses according to the Fokker-Planck equation [12]:

$$\frac{\partial p}{\partial t} = -\nabla_z \cdot ((Az + By)p) + \frac{1}{2} \text{Tr}(BQ_v B^T \nabla_z^2 p).$$

Even if v is non-Gaussian (leading to analytically intractable integro-differential diffusion), $z(t)$ becomes a continuous random variable.

Ultimately, the concurrent presence of topological indistinguishability and continuous stochastic diffusion demonstrates that the inverse mapping can no longer be treated as a simple function $x = T^\dagger(z)$ or even a deterministic set. It must instead be formulated as the problem of estimating the complex multimodal conditional distribution $p(x|z)$.

B. Set-Valued Estimation

When observability fails, T is no longer injective. The output history corresponds to a set of indistinguishable states:

$$\mathcal{X}^*(z) = \{x \in \mathcal{X} \mid T(x) = z\}. \quad (3)$$

We seek a set of estimates $\hat{\mathcal{X}}(t)$ wherein every generated point is physically consistent with the measured output. Mathematically, this requires the Hausdorff semi-distance $d_H^+(\hat{\mathcal{X}}(t), \mathcal{X}^*(z(t)))^1$ from the estimate set $\hat{\mathcal{X}}(t)$ to the true set $\mathcal{X}^*(z(t))$ to converge to zero. This ensures the estimates do not fall outside the valid physical set, even if they do not cover all possible modes. Within this general framework, we distinguish two fundamental scenarios:

- **Fixed Finite Cardinality:** In many indistinguishable physical systems, the set of indistinguishable states is restricted to a finite, fixed number of modes. This can be formalized as:

Assumption 1 (Bounded Cardinality). *There exists a constant integer $c \in \mathbb{N}^*$ such that, for all $z \in T(\mathcal{X})$, the number of pre-images is exactly c : $\text{card}(T^{-1}(z)) = c$.*

Under this assumption, we can strengthen our objective. Rather than an arbitrary set, we aim to produce exactly c distinct point estimates $\{\hat{x}^{(1)}(t), \dots, \hat{x}^{(c)}(t)\}$ that cover all possible states. In this case, the objective becomes convergence in the Hausdorff distance, guaranteeing that no valid physical mode is missed.

- **Smooth Trajectory Reconstruction:** Assuming the set-valued map $z \mapsto \mathcal{X}^*(z)$ is upper semi-continuous [13] and satisfies the regularity conditions of [11], the true indistinguishable states evolve continuously. The objective is therefore to reconstruct c continuous, non-chattering trajectories $\hat{x}^{(k)}(t)$ that smoothly track the physical modes over time without erratic jumping.

To jointly satisfy the aforementioned objectives, rather than relying solely on deterministic inversion, we aim to

¹The Hausdorff semi-distance from A to B is defined as $d_H^+(A, B) = \sup_{a \in A} \inf_{b \in B} \|a - b\|$.

characterize and approximate the full conditional probability distribution $p(x|z)$.

In the noiseless case ($v = 0$), the support of p captures the set $\mathcal{X}^*(z)$ and its distinct continuous trajectories. Specifically, when the cardinality is fixed to c , the distribution is a mixture of Dirac deltas, $p(x|z) = \sum_{k=1}^c w_k \delta(x - x^{(k)})$, where each $x^{(k)} \in \mathcal{X}^*(z)$ represents a distinct physical mode and w_k are the relative weights. Additionally, in the noisy case the variance of $p(x|z)$ quantifies the uncertainty induced by v .

Our approach provides a generative approximation of $p(x|z)$ via CFM, whose construction is detailed in Section III.

III. TRAINING METHOD

To approximate the set-valued inverse $\mathcal{X}^*(z)$, we aim to learn a generative map. Unlike regression models which map z to a single deterministic point (often yielding physically inconsistent averages in indistinguishable scenarios), we seek a mechanism that transforms a source of random noise into a sample from the conditional probability distribution $p(x|z)$, as illustrated in Figure 1.

This learning process relies on an offline training dataset $\mathcal{D} = \{(x, z)^{(k)}\}_{k=1}^{N_{\text{train}}}$, constructed by collecting pairs of true physical states x and their corresponding latent KKL states z [4], [5].

A. Conditional Flow Matching

The CFM paradigm is to learn a *vector field*, generating a *probability path* that transports a chosen noise distribution to the empirical data distribution defined by our dataset.

1) *Continuous Normalizing Flows:* The core idea is to define a continuous deformation of the probability space via an ODE evolving over a virtual time $\tau \in [0, 1]$. Let $\chi(\tau) \in \mathbb{R}^{n_x}$ be the state of a *particle* in this virtual flow. Its evolution is governed by a time-dependent vector field v_τ ² parameterized by the observer state z :

$$\frac{d\chi}{d\tau} = v_\tau(\chi, z), \quad \chi(0) = \chi_0 \sim \mathcal{N}(0, I_{n_x}) \quad (4)$$

where the initial condition is drawn from a Gaussian prior.

Our objective is to learn a vector field v_τ that transports the noise distribution to $p(x|z)$. Intuitively, integrating this ODE from $\tau = 0$ to $\tau = 1$ transforms an initial random noise sample χ_0 into a valid physical state.

2) *Per-Sample Paths:* Flow Matching builds the complex conditional distribution $p(x|z)$ from a mixture of simple, known deterministic trajectories. Instead of targeting the full distribution directly, consider a single target data point $x \in \mathbb{R}^{n_x}$ from the training set. We can explicitly define a *per-sample probability path* that transports a base Gaussian noise χ_0 precisely to this specific x .

While many paths exist, the *Optimal Transport* path [14] is the simplest, defined as a straight-line interpolation:

$$\chi = (1 - \tau)\chi_0 + \tau x. \quad (5)$$

²We use subscript to denote the time parameter τ .

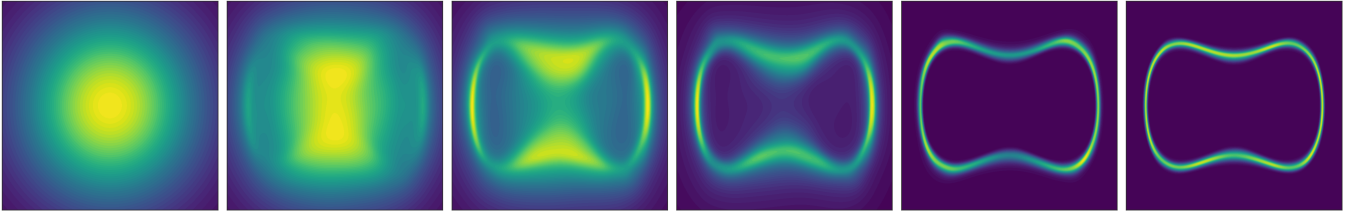


Fig. 1. Evolution of the conditional density distribution along the virtual time τ of the flow, for a fixed z . Each column corresponds to a snapshot at $\tau \in \{0, 0.02, 0.05, 0.1, 0.5, 1\}$. At $\tau = 1$, it corresponds to the approximation of $p(x|z)$.

By differentiating with respect to τ , we see that this path is generated by a constant per-sample vector field (which is independent of the guidance variable z):

$$u_\tau(\chi|x) = x - \chi_0. \quad (6)$$

3) *CFM Objective*: To generate the true conditional distribution $p(x|z)$, the overall marginal vector field $v_\tau(\chi, z)$ must aggregate the per-sample fields $u_\tau(\chi|x)$ associated with all possible valid states x . Mathematically, $v_\tau(\chi, z)$ is the expected value of $u_\tau(\chi|x)$ weighted by the posterior probability of the target x given the *current particle position* χ :

$$v_\tau(\chi, z) = \int u_\tau(\chi|x) p(x|\chi, z) dx. \quad (7)$$

This equation holds the key to why CFM does not collapse to a non-physical arithmetic mean. While standard regression only conditions on z , the CFM vector field is explicitly conditioned on the spatial position χ . As virtual time τ progresses, particles heading towards distinct physical modes become spatially separated. Once a particle χ gets closer to a specific mode $x^{(1)}$, the probability $p(x^{(1)}|\chi, z)$ approaches 1, and the local vector field strictly aligns with $u_\tau(\chi|x^{(1)})$, preventing non-physical averaging (as visually illustrated in Figure 2).

While computing (7) is intractable, the CFM framework [14] proves a fundamental equivalence: regressing against the per-sample fields u_τ implicitly minimizes the error against the field v_τ . We can thus parameterize the vector field with a neural network $v_\theta(\chi, \tau, z)$ and define a simulation-free loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau, \chi_0, (x, z)} \left[\|v_\theta(\chi, \tau, z) - (x - \chi_0)\|^2 \right], \quad (8)$$

where $\tau \sim \mathcal{U}[0, 1]$, $\chi_0 \sim \mathcal{N}(0, I_{n_x})$, and the pairs (x, z) are drawn from the dataset $\mathcal{D}$.

B. Computing $p(x|z)$

While forward integration of v_θ enables fast empirical *sampling* for our online estimator (Sec. IV), CFM also allows evaluating the exact density $p(x|z)$ for any x via the continuous normalizing flow formulation [15]. Integrating the coupled ODE backward from $\tau = 1$ to 0:

$$\begin{cases} \frac{d\chi}{d\tau} = v_\theta(\chi, \tau, z) \\ \frac{d \log p_\tau(\chi)}{d\tau} = -\nabla_\chi \cdot v_\theta(\chi, \tau, z) \end{cases} \quad (9)$$

with $\chi(1) = x$ yields the exact probability. The divergence $\nabla_\chi \cdot v_\theta$ is computed via automatic differentiation. Note that evaluating (9) is computationally demanding and used here strictly for offline visualization (e.g., Figure 4).

IV. ONLINE STATE ESTIMATION

With the trained vector field v_θ in hand, we now design the online estimation algorithm to fulfill the objectives defined in Section II-B.

A. Implementation: Clustering-and-Tracking

Given the latent trajectory $z(t)$ from the KKL filter, the estimation proceeds in two main phases: probabilistic generation (to reconstruct the set of indistinguishable states) and statistical aggregation (to extract usable trajectories).

1) *Initialization*: At $t = 0$, we sample N fixed initial noise vectors $\{\chi_0^{(i)}\}_{i=1}^N \sim \mathcal{N}(0, I_{n_x})$. These seeds are kept constant for all subsequent time steps to enforce temporal consistency. The hyperparameter N dictates the trade-off between computational cost and the empirical resolution of the generated distribution. In our experiments, $N = 100$ provided a satisfying balance.

2) *Set-Valued Generation via Flow Integration*: To address the *Set-Valued Indistinguishability* objective, we generate a particle cloud representing the conditional distribution $p(x|z(t))$. At any time t , this is achieved by solving the ODE (4) from $\tau = 0$ to $\tau = 1$ for each seed:

$$\chi^{(i)}(t) = \text{ODESolve}(v_\theta, \chi_0^{(i)}, z(t), \tau \in [0, 1]). \quad (10)$$

The resulting set $\{\chi^{(i)}(t)\}_{i=1}^N$ forms an empirical sample of the conditional distribution $p(x|z(t))$.

3) *Statistical Aggregation*: While the generated particle cloud captures the observability landscape, extracting specific continuous trajectories requires a structured statistical aggregation. To simultaneously address *Fixed Finite Cardinality*, and *Smooth Trajectory Reconstruction* objectives, we propose a *Clustering-and-Tracking* scheme operating in three steps:

a) *Partitioning (K-Means)*: First, to isolate the different modes of the distribution, we apply the K-Means algorithm to partition the N particles into c clusters (Figure 2), where c is the known cardinality of the indistinguishable set (for a strictly observable system, $c = 1$). Let $C_1, \dots, C_c$ denote the resulting particle clusters.

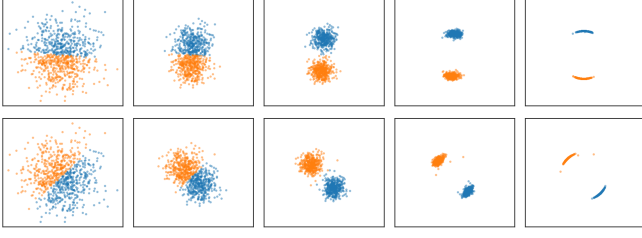


Fig. 2. Transport of samples $\chi_0 \sim \mathcal{N}(0, I)$ under the learned flow conditioned on the latent state z , shown at $\tau \in \{0.0, 0.5, 0.7, 0.9, 1\}$ (columns) and for two different z (rows). Colors indicate cluster assignment based on the final distribution at $\tau = 1$.

b) Robust Intra-Cluster Point Estimation (Geometric Median): Second, to avoid physically inconsistent states that can arise from standard arithmetic averaging within non-convex modes or in the presence of outliers, we extract a robust centroid for each cluster using the sample geometric median. For each cluster C_k , its representative μ_k is defined as the particle minimizing the intra-cluster distance:

$$\mu_k = \operatorname{argmin}_{x \in C_k} \sum_{\chi^{(i)} \in C_k} \|\chi^{(i)}(t) - x\|_2. \quad (11)$$

This yields a set of robust current estimates $\mathcal{C}_t = \{\mu_1, \dots, \mu_c\}$.

c) Temporal Tracking (Hungarian Algorithm): Finally, to ensure temporal continuity and prevent chattering between modes, we solve an optimal assignment problem between the current centroids $\mathcal{C}_t$ and the previous estimates $\hat{\mathcal{X}}(t^-) = \{\hat{x}^{(1)}(t^-), \dots, \hat{x}^{(c)}(t^-)\}$. Using the Hungarian algorithm, we find the permutation σ^* that minimizes the total squared displacement:

$$\sigma^* = \operatorname{argmin}_{\sigma \in S_c} \sum_{j=1}^c \|\hat{x}^{(j)}(t^-) - \mu_{\sigma(j)}\|^2. \quad (12)$$

The updated estimates $\hat{x}^{(j)}(t) = \mu_{\sigma^*(j)}$ allow each estimator to smoothly lock onto a specific physical branch without erratic jumping.

B. Theoretical Analysis

Let $\Phi_\theta(\chi_0, z)$ denote the solution of the Flow Matching ODE evaluated at the final virtual time $\tau = 1$:

$$\chi(t) = \Phi_\theta(\chi_0, z(t)) = \chi_0 + \int_0^1 v_\theta(\chi(\tau), \tau, z(t)) d\tau. \quad (13)$$

To ensure the observer's stability and regularity, we assume that the learned vector field v_θ is designed to be globally Lipschitz continuous with respect to both the state χ and the conditioning variable z . Note that v_θ is globally Lipschitz by construction, being a composition of bounded linear layers and globally Lipschitz activation functions. Under these assumptions, the KKL-CFM architecture satisfies the following properties.

a) Set-Valued Indistinguishability: In singular regions where the KKL immersion is non-injective, the generative nature of CFM allows for a continuous representation of the set of possible solutions.

Proposition 1 (Continuous Spatial Deformation). *Assume $v_\theta(\cdot, \tau, z)$ is globally Lipschitz continuous in space. For any fixed seed χ_0 , the generation map $(t, \chi_0) \mapsto \Phi_\theta(\chi_0, z(t))$ is jointly continuous.*

Proof. Using the Cauchy-Lipschitz theorem, we show that the flow $\chi_0 \mapsto \Phi_\theta(\chi_0, z)$ is a continuous bijection. Since the observer state $z(t)$ evolves continuously, the joint mapping remains continuous. $\square$

Remark 1. *While the true underlying mapping separating distinct physical modes inherently contains a spatial discontinuity (a separatrix), the neural network parameterization regularizes it into a globally Lipschitz vector field v_θ . Thus, the trajectory of any single initial condition χ_0 is continuous. However, this regularized separatrix retains a large local Lipschitz constant. When it sweeps across a particle (Figure 2), the state undergoes a numerically “stiff”, yet mathematically continuous, transition.*

Our Clustering-and-Tracking aggregation (Section IV-A3) mitigates this effect. Because the separatrix moves continuously, the fraction of particles crossing it at any instant is minimal. By averaging particles via K-Means, the impact of these individual jumps is attenuated by $\mathcal{O}(1/N)$.

b) Smooth Trajectory Reconstruction:

Proposition 2 (Temporal Regularity). *Assuming the measurement $y(t)$ is continuous, the individual trajectory $\chi^{(i)}(t) = \Phi_\theta(\chi_0^{(i)}, z(t))$ is continuously differentiable with respect to t .*

Proof. Since v_θ is $\mathcal{C}^1$ with respect to z , the flow map Φ_θ is $\mathcal{C}^1$. Its composition with the $\mathcal{C}^1$ KKL dynamics $\dot{z}(t) = Az(t) + By(t)$ yields a continuously differentiable physical trajectory. $\square$

Remark 2. *This result ensures the online estimator is structurally well-posed and deterministic. It does not, however, address the statistical convergence of the learned distribution to the ground truth, which remains an open challenge in deep generative modeling.*

c) Robustness to Measurement Noise:

Proposition 3 (Lipschitz Inversion). *Let L_x and L_z be the Lipschitz constants of v_θ w.r.t. x and z respectively. For a fixed χ_0 , the inverse mapping $z \mapsto \Phi_\theta(\chi_0, z)$ is globally Lipschitz continuous.*

Proof. Let z_1, z_2 be two observer states, and $\chi_1(\tau), \chi_2(\tau)$ their corresponding flows. We have $\frac{d}{d\tau} \|\chi_1 - \chi_2\| \leq L_x \|\chi_1 - \chi_2\| + L_z \|z_1 - z_2\|$. Applying Grönwall's inequality gives:

$$\|\Phi_\theta(\chi_0, z_1) - \Phi_\theta(\chi_0, z_2)\| \leq L_z \frac{e^{L_x} - 1}{L_x} \|z_1 - z_2\|. \quad (14)$$

$\square$

Consequently, bounded perturbations in the latent space z (induced by measurement noise y) result in strictly bounded estimation errors, inheriting the Input-to-State Stability of the CFM-KKL.

V. NUMERICAL EXPERIMENTS

To demonstrate the practical efficacy of the KKL-CFM architecture, we evaluate its performance against the estimation objectives defined in Section II.

A. Experimental Setup, Benchmarks & Evaluation

For all experiments, we use a KKL observer with a latent dimension $n_z = n_y(2n_x + 1)$ to ensure immersion properties. Following the tuning strategy proposed in [5], the dynamics matrix A is designed as a Butterworth filter. The generative vector field $v_\theta(\chi, \tau, z)$ is parameterized by a Multi-Layer Perceptron (MLP) with 4 hidden layers of 128 units and GeLU activations. During inference, the ODEs for both particle generation and conditional density $p(x|z)$ computation are integrated using a standard RK4 scheme. For noisy scenarios, the output is corrupted by zero-mean Gaussian noise with standard deviation σ_v . Implementation details and source code are provided at <https://github.com/jolindien-git/DeepKKL>.

Metrics: We quantify performance using the time-averaged directed Modified Hausdorff Distance (MHD) between the ground-truth set $\mathcal{X}^*(t)$ and the estimated set $\hat{\mathcal{X}}(t)$. The final scores are obtained by averaging the following instantaneous distances over the sequence length. We evaluate two axes:

- *Precision* measures adherence to the true manifold (penalizing spurious estimates):

$$d_{\text{prec}}(t) = \frac{1}{|\hat{\mathcal{X}}(t)|} \sum_{\hat{x} \in \hat{\mathcal{X}}(t)} \inf_{x^* \in \mathcal{X}^*(t)} \|\hat{x} - x^*\| \quad (15)$$

- *Coverage* measures exhaustiveness across all true branches (penalizing mode collapse):

$$d_{\text{cov}}(t) = \frac{1}{|\mathcal{X}^*(t)|} \sum_{x^* \in \mathcal{X}^*(t)} \inf_{\hat{x} \in \hat{\mathcal{X}}(t)} \|\hat{x} - x^*\| \quad (16)$$

Baselines: To evaluate our proposed Generative KKL (KKL-CFM), we benchmark it against three distinct state-of-the-art methodologies. The *Neural-ODE KKL* [16] serves as a standard global deterministic baseline, utilizing neural ordinary differential equations to approximate a single-valued inverse mapping. The *Set-Valued KKL* [17], a theoretical extension of the standard KKL observer that formulates a set-valued inversion map. In algorithmic practice, however, its optimization-based inversion systematically converges to a single local minimizer, failing to capture the multiplicity of the indistinguishable set. Finally, the *Bootstrap Particle Filter (BPF)* provides a purely stochastic baseline, propagating particles to approximate the posterior distribution. To enable a fair comparison, the BPF was equipped with our proposed Clustering-and-Tracking scheme to extract continuous multimodal trajectories.

B. Finite Cardinality and Smooth Trajectory Tracking

We evaluate the tracking capabilities of the proposed architecture on three nonlinear systems exhibiting varying degrees of indistinguishability.

The first system is the standard *Van der Pol oscillator* [4], [16], which serves as a baseline for unimodal estimation where the standard KKL observer is naturally injective.

The second is a *Bimodal System* [11], [17]:

$$\begin{aligned} f(x) &= [x_2 + x_1(1 - (x_1^2 + x_2^2)), -x_1 + x_2(1 - (x_1^2 + x_2^2))]^\top, \\ h(x) &= [x_1^2 - x_2^2, 2x_1x_2]^\top. \end{aligned} \quad (17)$$

If $x(t)$ is a solution, then $-x(t)$ is also a valid solution producing the exact same output $y(t)$. The backward-indistinguishable set here has a finite cardinality of 2, i.e., $|\mathcal{X}^*| = 2$.

We next introduce a new benchmark consisting of two coupled *Van der Pol oscillators*:

$$\begin{aligned} f(x) &= [x_2, \mu(1 - x_1^2)x_2 - \omega_1^2x_1 + k(x_3 - x_1), \\ &\quad x_4, \mu(1 - x_3^2)x_4 - \omega_2^2x_3 + k(x_1 - x_3)]^\top, \\ h(x) &= [x_1^2, x_3^2]^\top, \end{aligned} \quad (18)$$

with parameters $\mu = 1.0$, $k = 0.1$, $\omega_1 = 1.0$, and $\omega_2 = \sqrt{2}$. This configuration creates four symmetrically indistinguishable orbits ($|\mathcal{X}^*| = 4$).

System	Method	$\sigma_v = 0$		$\sigma_v = 1$	
		Prec.	Cov.	Prec.	Cov.
Van der Pol $ \mathcal{X}^* = 1$	Neural-ODE KKL	0.0069	0.0069	0.17	0.17
	Set-Valued KKL	0.025	0.025	0.45	0.45
	BPF + Track	0.23	0.23	0.61	0.61
	Generative KKL	<u>0.0076</u>	<u>0.0076</u>	0.17	0.17
Bimodal $ \mathcal{X}^* = 2$	Neural-ODE KKL	1.0	1.0	1.0	1.0
	Set-Valued KKL	<u>0.0049</u>	1.0	0.28	1.2
	BPF + Track	0.56	<u>0.74</u>	<u>0.14</u>	<u>0.18</u>
	Generative KKL	0.0030	0.0030	0.084	0.084
Coupled Vand der Pol $ \mathcal{X}^* = 4$	Neural-ODE KKL	3.2	3.2	3.2	3.2
	Set-Valued KKL	<u>1.6</u>	4.0	<u>1.8</u>	4.1
	BPF + Track	2.5	<u>2.7</u>	2.2	2.5
	Generative KKL	0.42	0.43	0.39	0.39

TABLE I

PERFORMANCE COMPARISON ACROSS DIFFERENT SYSTEMS AND MEASUREMENT NOISE LEVELS (σ_v). THE ABBREVIATIONS “PREC.” AND “COV.” RESPECTIVELY DENOTE THE ESTIMATION PRECISION (15) AND ESTIMATION COVERAGE (16) METRICS. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED.

The results presented in Table I highlight several fundamental differences between the evaluated methods. As expected, the *Neural-ODE KKL* performs well on the classical VDP system, which is unimodal. However, it fails on the multimodal systems. The *Set-Valued KKL* observer strong precision scores, validate its ability to isolate valid modes. Nevertheless, it also fails to maintain multiple hypotheses and struggles in the presence of measurement noise ($\sigma_v = 1$). The *BPF + Tracking* approach maintains reasonable performance, though it paradoxically performs slightly worse in the noiseless scenario ($\sigma_v = 0$), an issue caused by particle weight degeneracy.

Finally, our proposed *Generative KKL* (KKL-CFM) remains competitive across all configurations, bridging the gap between discrete multimodality and continuous noise.

The competitive metrics achieved by our *Generative KKL* on the multimodal systems is illustrated by Fig. 3. It details the temporal tracking for the bimodal system. As shown it extracts the two distinct trajectories $\hat{x}^{(0)}$ and $\hat{x}^{(1)}$, allowing each estimator to continuously lock onto its respective physical branch without chattering.

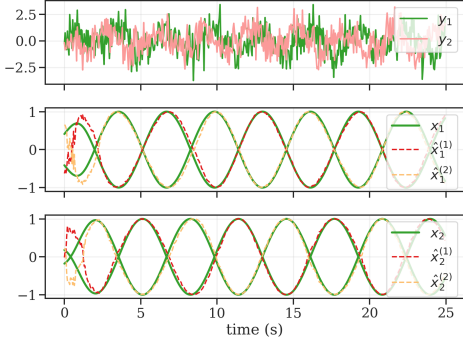


Fig. 3. Continuous tracking of the two indistinguishable solutions of system (17) by KKL-CFM under measurement noise ($\sigma_v = 1$).

C. Infinite Cardinality and Set-Valued Estimation

We finally consider the undamped Duffing oscillator [18]:

$$\begin{aligned} f(x) &= [x_2, x_1 - x_1^3]^\top, \\ h(x) &= -\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{4}x_1^4. \end{aligned} \quad (19)$$

The nominal system is Hamiltonian. For a given initial condition x_0 , the the output $h(x)$ remains strictly constant over time. Consequently, the backward-indistinguishable set is not a finite collection of points, but an infinite-cardinality set (a continuous 1D curve corresponding to the entire energy orbit).

Fig. 4 presents the conditional density $p(x|z)$ estimated by KKL-CFM for two distinct initial energy levels. The method successfully reconstructs the continuous non-convex physical sets: the probability mass is distributed along the continuous orbits. This confirms that the latent state z successfully encodes the dynamics of the specific energy level, and the Flow Matching decoder reliably maps this representation back to the infinite set of valid physical states.

VI. CONCLUSION

This paper presented KKL-CFM, a generative architecture for nonlinear state estimation that addresses the inherent limitations of deterministic KKL inversion. By approximating the conditional state distribution, our approach bypasses injectivity issues in singular regions and filters noise via a Clustering-and-Tracking scheme.

A key limitation occurs when indistinguishable trajectories evolve spatially close. If their separation matches the particle cloud variance, K-means may fail to partition modes, causing chattering. Future work will integrate temporal priors to improve tracking consistency in such overlapping regions.

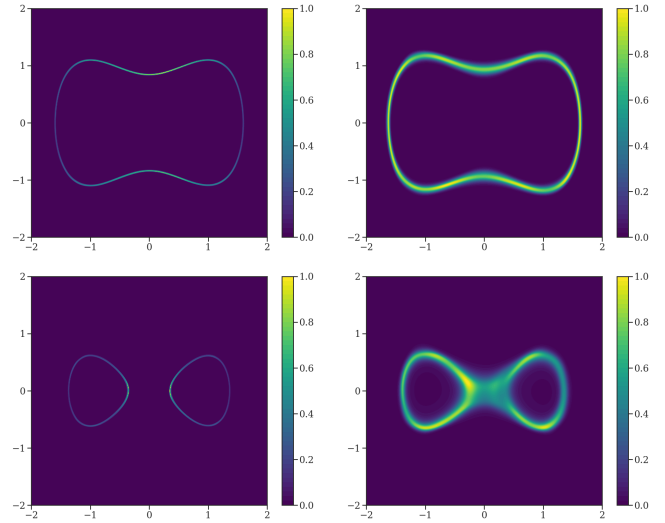


Fig. 4. Estimated conditional density $p(x|z)$ for the Duffing system, for two distinct orbits (rows). The generative flow recovers the infinite-cardinality indistinguishable set $\mathcal{X}^*(z)$, corresponding to a continuous energy level. *Left*: noise-free measurements ($\sigma_v = 0$). *Right*: noisy measurements ($\sigma_v = 1$).

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G. APPENDIX: NUMBER OF PARTICLES

To further evaluate the computational efficiency and scalability of our approach, we conduct an ablation study on the number of generated particles N used during the online estimation phase. We compare the performance of our generative KKL (KKL-CFM) against the Bootstrap Particle Filter (BPF) baseline under a high measurement noise level ($\sigma_v = 1$) across the three benchmark systems: *Van der Pol*, *Bimodal*, and *Coupled Van der Pol*. For both methods, we vary the number of particles N logarithmically, and report the resulting tracking Precision and Coverage errors.

The results, illustrated in Figure 5, reveal a stark contrast in sample efficiency between the two methods. The performance of the BPF degrades severely as N decreases, particularly with the *Coupled Van der Pol* system. This sensitivity directly illustrates the curse of dimensionality inherent to classical particle filtering: the number of samples required to adequately represent complex, distributions grows prohibitively with the system’s complexity. This behavior is a direct consequence of particle weight degeneracy (Section H-B). In contrast, KKL-CFM demonstrates remarkable stability with respect to the sample size. Because the learned vector field provides a deterministic and structurally continuous spatial deformation, KKL-CFM achieves low and consistent tracking errors even with as few as $N = 16$ particles. This confirms that our generative architecture not only accurately reconstructs multimodal distributions, but does so with a significantly reduced computational burden, making it highly suitable for real-time online deployment.

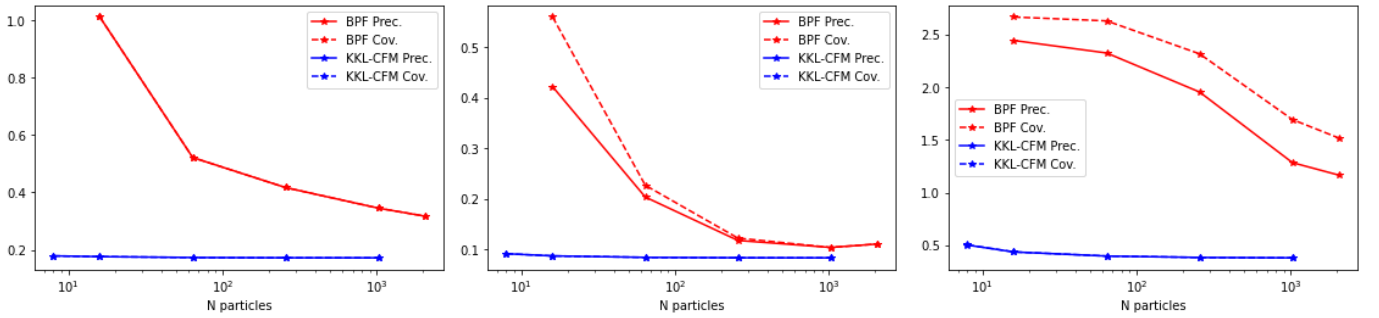


Fig. 5. Results vs Number of particles (noise level: $\sigma_v = 1$). (Left): Van der Pol. (Middle): BiModal. (Right): Coupled Van der Pol.

H. APPENDIX: PARTICLE FILTER (BASELINE)

A. Algorithm Overview

To handle the indistinguishability of nonlinear systems, we could resort to a Bootstrap Particle Filter (BPF). The BPF approximates the posterior distribution $p(x(t) | y_{0:t})$ of the physical state given noisy measurements, using a set of N_{part} discrete particles $\{\hat{x}^{(i)}\}_{i=1}^{N_{\text{part}}}$ with associated normalized weights $\{w_i\}_{i=1}^{N_{\text{part}}}$.

The BPF algorithm proceeds in three recursive steps at each time increment δ :

- 1) **Prediction:** Each particle is propagated forward in time by integrating the continuous system dynamics over the interval δ . For deterministic systems, an artificial roughening noise $\eta^{(i)}$ is injected to prevent sample impoverishment:

$$\hat{x}^{(i)}(t + \delta) = \hat{x}^{(i)}(t) + \int_t^{t+\delta} f(\hat{x}^{(i)}(s))ds + \eta^{(i)}(t), \quad \eta^{(i)}(t) \sim \mathcal{N}(0, Q_r) \quad (20)$$

where Q_r is the chosen roughening covariance matrix.

- 2) **Update:** Upon receiving a new measurement $y(t)$, the weights are updated based on the likelihood of the observation. Assuming zero-mean Gaussian measurement noise with covariance R , the recursion is:

$$w^{(i)}(t) \propto w^{(i)}(t - \delta) \cdot \exp\left(-\frac{1}{2}\|y(t) - h(\hat{x}^{(i)}(t))\|_{R^{-1}}^2\right) \quad (21)$$

The weights are then normalized such that $\sum_{i=1}^{N_{\text{part}}} w^{(i)}(t) = 1$.

- 3) **Resampling:** To prevent the degeneracy where all but one particle have negligible weights, a systematic resampling step is performed whenever the effective sample size $N_{\text{eff}} = (\sum w^{(i)}(t)^2)^{-1}$ falls below a threshold (typically $N_{\text{part}}/2$).

B. Limitations: Mode Collapse and Particle Depletion

A significant challenge in using Particle Filters for multimodal estimation is the phenomenon of **Mode Collapse** (or Sample Impoverishment). In scenarios with very low measurement noise ($R \ll 1$), the likelihood function becomes extremely peaked. During the resampling stage, particles belonging to a mode that is even slightly “less probable” due to stochastic noise may be entirely discarded in favor of particles from a more dominant mode.

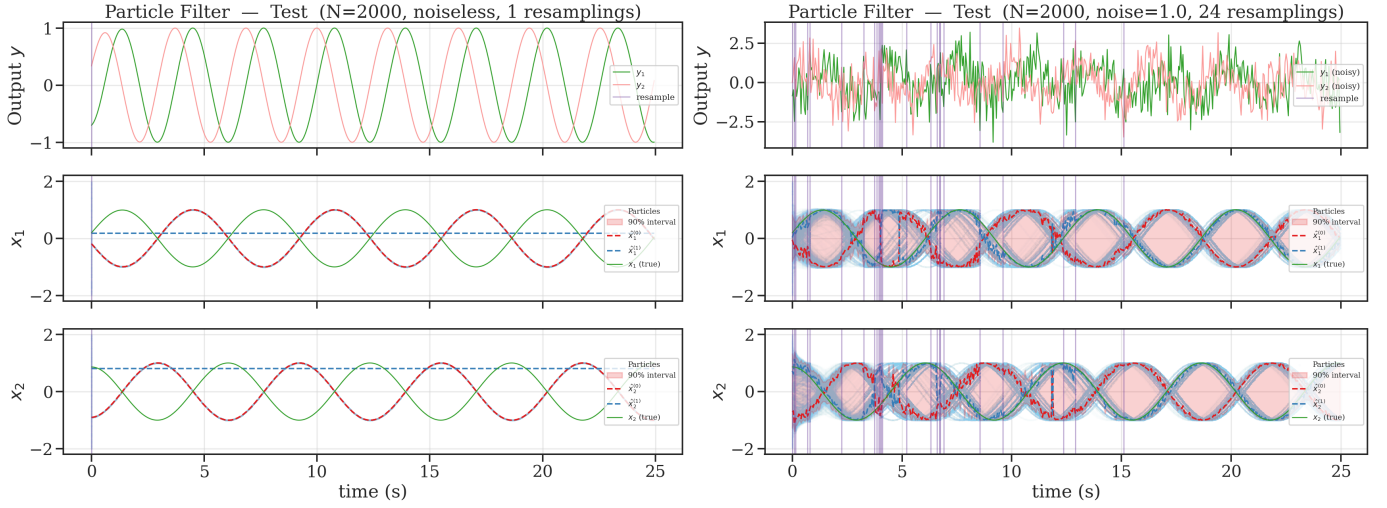


Fig. 6. Particle filter with K-means for the Test dataset. Left: noiseless case. Right: noise std of 1.0

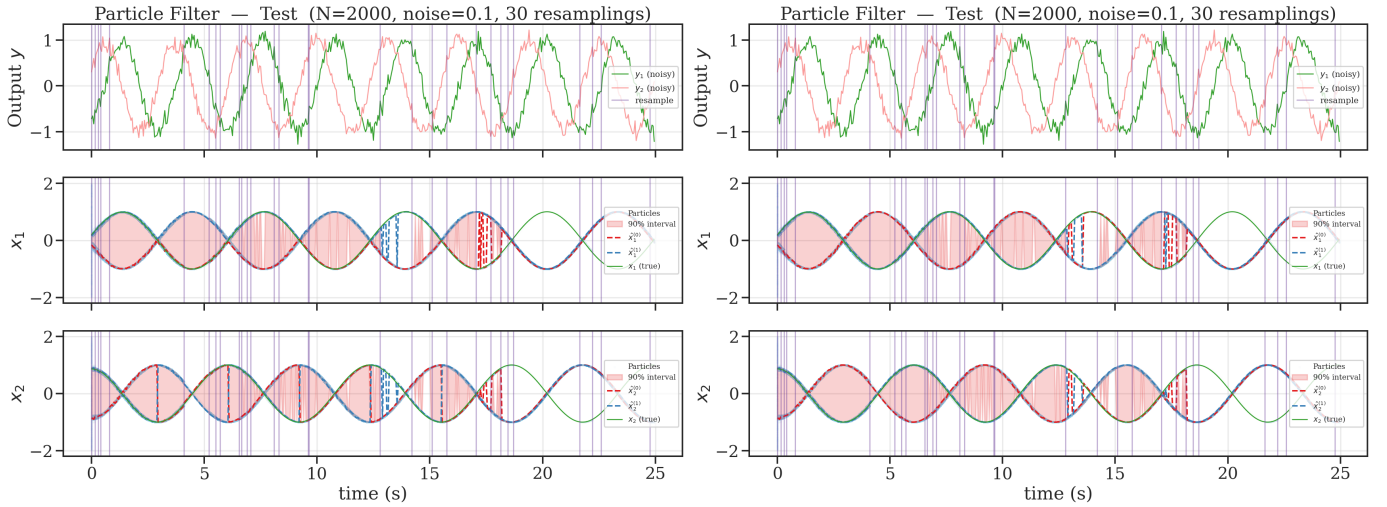


Fig. 7. Particle filter with K-means for the Test dataset with noise std of 0.1. Left: only K-means, no info on previous centroids. Right: K-means + Hungarian algorithm (with info on the previous centroid labels).

This leads to *particle depletion*, where the filter loses its ability to represent the full set-valued nature of the observer, eventually collapsing into a unimodal distribution. This can be mitigated through heuristics such as:

- **Measurement Noise Floor:** Artificially increasing R in the filter to broaden the likelihood.
- **Roughening/Jittering:** Injecting additional noise post-resampling to force particle diversity.
- **Prior Injection:** Initializing a small percentage of particles at each step from the prior distribution.

However, it remains an intrinsic limitation of Sequential Monte Carlo methods. The trade-off between tracking accuracy (high confidence) and multi-hypothesis maintenance (high diversity) is fundamental, as the resampling step is designed to minimize variance by focusing on the most likely regions of the state space, often at the expense of secondary modes.

The simulation results shown in Figure 6 and Figure 7 illustrate the trade-offs between tracking precision and the maintenance of a set-valued estimate.

Sensitivity to Measurement Noise: In the noiseless case ($\sigma = 0.0$), the particle filter exhibits immediate mode collapse. The infinitely peaked likelihood function ensures that even infinitesimal differences in particle weights lead to the total removal of one of the modes during resampling. Conversely, at $\sigma = 1.0$, the broadened likelihood allows for the simultaneous tracking of both indistinguishable modes, albeit at the cost of significantly higher estimation variance and wider probability bands.

Temporal Stability and Depletion: As shown in Figure 7 for the $\sigma = 0.1$ case, raw k -means clustering is insufficient for continuous estimation due to frequent label-switching. While the integration of the Hungarian algorithm stabilizes these trajectories, it does not prevent eventual particle depletion. The variance-reduction property of the resampling step tends to concentrate particles into a single mode over long horizons, effectively collapsing the set-valued observer into a unimodal

estimator. This highlights that while heuristics like label-tracking provide short-term stability, the maintenance of multimodal diversity remains a fundamental challenge in BPF.

I. STOCHASTIC ANALYSIS OF THE KKL OBSERVER

A. Presence of Measurement Noise

Consider the nonlinear system subject only to measurement noise:

$$\dot{x} = f(x), \quad y = h(x) + v(t) \quad (22)$$

where $v(t)$ is modeled as a Gaussian white noise with covariance Q_v .

1) *Uncertainty on z* : The observer's differential equation becomes an Itô Stochastic Differential Equation (SDE):

$$d\hat{z} = (A\hat{z} + Bh(x))dt + B\sqrt{Q_v}dW_t \quad (23)$$

where W_t is a standard Wiener process (Brownian motion).

The exact time evolution of the probability density function $p(\hat{z}, t)$ is governed by the Fokker-Planck equation:

$$\frac{\partial p}{\partial t} = -\nabla_{\hat{z}} \cdot \left((A\hat{z} + Bh(x))p \right) + \frac{1}{2} \text{Tr} \left(BQ_v B^T \nabla_{\hat{z}}^2 p \right) \quad (24)$$

The first term (drift) pushes the density to follow the nominal dynamics of the observer. The second term (diffusion) spreads the density due to the measurement noise entering through matrix B .

If the initial belief $p(\hat{z}_0)$ is Gaussian (or a Dirac delta) and the physical trajectory $x(t)$ is deterministic and fixed, then the conditional distribution $p(\hat{z}, t \mid x_{0..t})$ remains a Gaussian for all time t :

$$p(\hat{z}, t \mid x_{0..t}) = \mathcal{N}(\mu_z(t), \Sigma_z(t)), \quad (25)$$

where:

- The mean μ_z exactly follows the nominal (noise-free) observer dynamics:

$$\dot{\mu}_z = A\mu_z + Bh(x). \quad (26)$$

The expected value of the observer thus converges to the ideal trajectory $z = T(x)$.

- The covariance Σ_z measures the uncertainty (spread) around the mean, and evolves according to the continuous Lyapunov equation:

$$\dot{\Sigma}_z = A\Sigma_z + \Sigma_z A^T + BQ_v B^T \quad (27)$$

Since the matrix A is Hurwitz, the covariance equation is stable. As $t \rightarrow \infty$, the covariance tends to a constant limit Σ_∞ , which is the unique solution to the Algebraic Lyapunov Equation:

$$A\Sigma_\infty + \Sigma_\infty A^T + BQ_v B^T = 0. \quad (28)$$

Remark (Uncertainty on the initial state x_0): The Gaussian property of $\hat{z}$ described above strictly holds conditionally on a known deterministic trajectory $x(t)$. If we consider a more realistic scenario where the initial physical state is uncertain, e.g., $x_0 \sim \mathcal{N}(\mu_{x_0}, \Sigma_{x_0})$, the true trajectory $x(t)$ becomes a stochastic process. Due to the nonlinearities in both the system dynamics $f(x)$ and the observation function $h(x)$, the distribution of the forcing term $h(x(t))$ becomes highly non-Gaussian. Consequently, the marginal distribution $p(\hat{z}, t)$ loses its Gaussian nature and becomes analytically intractable, further motivating the need for expressive data-driven generative models.

2) *Uncertainty on x* : The total variance of the state estimation x can be decomposed using the law of total variance:

$$\Sigma_x = \mathbb{E}_{\hat{z}}[\text{Var}(x|\hat{z})] + \text{Var}_{\hat{z}}(\mathbb{E}[x|\hat{z}]), \quad (29)$$

where:

- $\mathbb{E}_{\hat{z}}[\text{Var}(x|\hat{z})]$ represents the intrinsic uncertainty of the system, inherently linked to indistinguishability.
- $\text{Var}_{\hat{z}}(\mathbb{E}[x|\hat{z}])$ represents the propagation of the measurement noise through the inverse mapping.

B. Presence of Process and Measurement Noise

Consider the fully uncertain nonlinear system:

$$\dot{x} = f(x) + w(t), \quad y = h(x) + v(t) \quad (30)$$

where $w(t) \sim \mathcal{N}(0, Q_w)$ and $v(t) \sim \mathcal{N}(0, Q_v)$ are assumed to be independent white Gaussian noises.

1) *Joint Uncertainty and Intractability*: Because the physical state x_t is now stochastic due to the process noise $w(t)$, the observer state $\hat{z}_t$ alone is no longer a Markov process. We must define the joint SDE:

$$\begin{cases} dx_t = f(x_t)dt + \sqrt{Q_w}dW_{w,t} \\ d\hat{z}_t = (A\hat{z}_t + Bh(x_t))dt + B\sqrt{Q_v}dW_{v,t} \end{cases} \quad (31)$$

The evolution of the system is governed by the joint Fokker-Planck equation:

$$\frac{\partial p}{\partial t} = \underbrace{-\nabla_x \cdot (f(x)p) - \nabla_{\hat{z}} \cdot ((A\hat{z} + Bh(x))p)}_{\text{Drift terms}} + \underbrace{\frac{1}{2}\text{Tr}(Q_w \nabla_x^2 p) + \frac{1}{2}\text{Tr}(BQ_v B^T \nabla_{\hat{z}}^2 p)}_{\text{Diffusion terms}} \quad (32)$$

Due to the nonlinearities in $f(x)$ and $h(x)$, this joint partial differential equation has no general closed-form analytical solution. Furthermore, the marginalization required to obtain $p(\hat{z}, t)$ or the posterior $p(x|\hat{z})$ is computationally intractable. This analytical bottleneck justifies shifting the computational burden offline by leveraging data-driven generative models, such as Conditional Flow Matching, to approximate the complex posterior distribution.